Mechanisms of damage

1. Organisms enter through the skin, multiply there and produce the disease in the skin.
   For example, impetigo, abscess and cellulitis (Figure 12.7) are caused by *Staphylococcus aureus* and *Streptococcus pyogenes*.



Figure 12.7: Cellulitis

As soon as the organisms enter the skin they multiply and produce various toxins that kill the cells and produce cellulitis. Further damage leads to necrosis and ulcer formation (Figure 12.8).



Figure 12.8: Ulcer formation

2. Organisms multiply in the skin and produce disease in internal organs. For example some Group A *Streptococci* multiply in the skin and produce disease known as Acute Glomerulonephritis causing damage to the kidneys. Some times *Corynebacterium diphtheriae* may multiply in the skin and affect the heart due to the toxin

Table 12.2: Bacterial Infections of the skin

Disease	Pathogen	Signs and Symptoms	Transmission
Cellulitis	*Streptococcus pyogenes*	Localised inflammation of dermis and hypodermis; skin red, warm, and painful to the touch	Through cut or abrasion
Erysipelas	*Streptococcus pyogenes*	Inflamed, swollen patch of skin, often on face; may be suppurative	Through cut or abrasion
Impetigo	*Staphylococcus aureus, Streptococcus pyogenes*	Vesicles, pustules, and sometimes bullae around nose and mouth	Highly contagious, especially via contact
Wound infections	*Pseudomonas aeruginosa,* others	Formation of biofilm in or on wound	Exposure of wound to microbes in environment; poor wound hygiene

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